Beyond these traditional concerns, a growing number of activists have joined the cause of safeguarding the space environment. They recognise the importance of preserving the pristine nature of space and preventing the proliferation of space debris. As satellites become more ubiquitous and constellations grow in number, these activists advocate for responsible satellite deployment and the adoption of measures to mitigate the potential environmental impact.

Amidst this backdrop of interconnectedness, vulnerabilities, and concerns, space cyber security emerges as a captivating field of study. It delves into the intricacies of data security within transmission networks that facilitate satellite communication. It encompasses a wide range of aspects, from the signal processing mechanisms employed in the control segments to the security of orbital objects and their complex onboard systems.

In essence, space cyber security fuses the domains of technology and protection, weaving together the realms of satellite operations and cybernetics. It represents a captivating intersection where experts explore innovative solutions to secure the transmission of data and fortify the resilience of satellites against cyber threats in our ever-expanding digital universe.

Challenges and innovations in Low Earth Orbit

Within the realm of commercial space applications, a significant focus lies on activities taking place in Low Earth Orbit (LEO). This primarily involves the utilisation of remote sensing satellites for Earth observation and monitoring human activities. While a smaller portion of LEO activities is dedicated to communication services, there has been a notable rise in NewSpace applications that allow for the development and deployment of satellite configurations such as swarms, CubeSats, and nanosatellites. These innovations offer cost-effective solutions for Earth observation and integrated applications, albeit with a trade-off in terms of reduced reliability.

Of particular interest are the constellations of satellites that are being deployed in LEO. These constellations consist of a large number of interconnected satellites, with some comprising thousands of individual assets. While these constellations bring benefits such as enhanced coverage and connectivity, they also present significant challenges. One such challenge is collision avoidance. With a large number of satellites occupying the same orbital space, the risk of collision increases, necessitating the implementation of robust collision avoidance systems and effective space traffic management frameworks.

One incident that exemplifies this risk occurred in April 2020 when two satellites from the Starlink and OneWeb constellations came dangerously close to each other. This event underscored the pressing need for collision avoidance measures and highlighted the importance of developing a comprehensive space traffic management framework.

Balancing latency and coverage

One of the advantages of utilising LEO for satellite operations is the reduced communication latency compared to satellites in Geostationary Earth Orbit (GEO). This low latency is particularly advantageous for applications such as global 5G internet services, providing a competitive edge in markets reliant on fast and responsive internet connectivity. However, due to the nature of LEO orbits, a single satellite can only cover a small portion of the Earth's surface within a short time window. As a result, more satellites are required in a constellation to achieve global coverage compared to satellites in GEO. Managing the network of interconnected satellites in LEO becomes complex due to the constantly changing topology and high mobility, posing challenges for reliable information transmission between ground stations, satellites, and satellite-to-satellite links.

The proliferation of satellites within large constellations contributes to congestion in LEO, exacerbating the risk of collisions. Furthermore, as projections indicate, a significant percentage of satellites within these constellations may become inoperable over time, leading to the generation of space debris. The issue of space debris is a growing concern for LEO, especially with regard to the sustainability and long-term viability of space operations. Effective space debris mitigation strategies and policies are necessary to address this risk and ensure a safe and sustainable space environment.

LEO also poses challenges in terms of satellite lifetime and disposal. Satellites in LEO can remain in orbit for several decades after their operational service ends, depending on factors such as atmospheric drag. Proper disposal mechanisms, such as using drag devices for self-deorbiting, are crucial to prevent inactive satellites from becoming hazards that can potentially collide with operational assets. Integrating deorbiting plans into satellite designs can help mitigate risks and ensure responsible end-of-life management.

Additionally, the use of CubeSats and nanosatellites introduces new complications. These smaller assets, while enabling innovative and cost-effective space missions, are often challenging to track and monitor